



NATIONAL OPEN UNIVERSITY OF NIGERIA
University Village, Plot 91, Cadastral Zone, Nnamdi Azikwe Express Way, Jabi-Abuja

FACULTY OF SCIENCES
DEPARTMENT OF MATHEMATICS
2024_2 EXAMINATION

Course Code: MTH301

Credit Unit: 3

Total: 70 Marks

Course Title: Functional Analysis

Time Allowed: 3 Hours

Instruction: Answer Question One (1) and Any Other Three (3) Questions

Q1(a)i) Define each of the following:

(i) Discrete topology

(3 marks)

(ii) Indiscrete topology.

(2 marks)

(b) i) Define a metric d on a set X .

(5 marks)

ii) Suppose $\{x_n\}$ is a sequence in a metric space (X, d) . Show that there exists at most one point $x \in X$ such that $\{x_n\}$ converges to x .

(7 marks)

(c) If (X, d) is a complete metric and Y is a closed subspace of X . Establish that (Y, d) .

(5 marks)

Q2 (a) Define neighborhood of x **(3 marks)** ii) subsequence of $\{x_n\}$ **(4marks)**

(b) Established that a subset A of X is open if and only if its complement A^c is closed in X .

(9 marks)

Q3 (a) Define a sequence $\{f_n\}$ that converges uniformly to f .

(4 marks)

(b) Let $\{f_n\}$ be a sequence of continuous functions from a metric space (X, d) to a metric

(Y, ρ) . Suppose that $\{f_n\}$ converges to f from X to Y . Show that f is continuous. **(12 marks)**

Q4 (a) Define a compact metric space. When is a subspace said to be compact? **(4 marks)**

(b) State two (2) elementary properties of continuous functions between topological spaces.

(6 marks)

(c) Suppose (X, d) is compact and Y is a closed subset of X . Establish that Y is compact.

(6 marks)

Q5 (a) Define each of the following:

(i) a separation **(4 marks)**

(ii) a connected subset Y of X

(4 marks)

(b) Differentiate between connected and disconnected topological spaces.

(3 marks)

(c) Show that if $\{A_i\}_{i \in I}$ is a family of connected subsets of X such that $\cap_{i \in I} A_i \neq \emptyset$, then

$A = \cup_{i \in I} A_i$ is connected.

(5marks)